

HEIMDALL

Smart Campus Access & Hostel Management System

END-TO-END PORTALS GUIDE

A comprehensive manual detailing operations for Students, Guards, Wardens, and Admins.

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1. Executive Overview

Heimdall is a high-availability campus gate control system designed to streamline permission checking, outpasses, and student movement tracking using dynamic QR codes. By integrating real-time Socket.io dashboards, parent-approval WhatsApp automation (via BullMQ queues), and a multi-level role authorization schema, Heimdall bridges the security gap between students, security guards, wardens, and administrative headers.

Core Portals & Workflows Covered in This Guide:

- **1. Student Portal:** Self-service dashboard to generate QR gate-passes, track active status, file complaints, and request parent-approved home visits.
- **2. Guard Portal:** Hardware-integrated camera scanning page with immediate gate clearance validation, status alerts, and access beep indicators.
- **3. Warden Portal:** Operational hub for auditing out-of-campus logs, managing complaints, and performing final approvals on home visits.
- **4. Admin Portal:** Telemetry and system health console showing connection status, resource charts, API performance, and config stats.

2. The Student Portal

The Student Portal acts as the central interface for students living in the campus hostels. The interface uses a dark glassmorphic design system that ensures optimal legibility on both mobile screens (at the gates) and desktop environments.

Key Functions and Step-by-Step Flow:

- **Dynamic QR Generation:** Every student gets a unique QR code generated using their encrypted token mixed with a time-based OTP salt. QR codes automatically expire to prevent screenshot sharing or duplicate entries.
- **Quick Status Indicator:** Displays in real-time whether the student is marked "IN HOSTEL" (green) or "Checked OUT" (orange).
- **Home Visit Pass Request:** Students submit outpass dates and destinations. This triggers an automated WhatsApp permission SMS to the parent.
- **Real-Time Logging:** Maintains a live, audit-ready log of the student's individual past entries and exits.

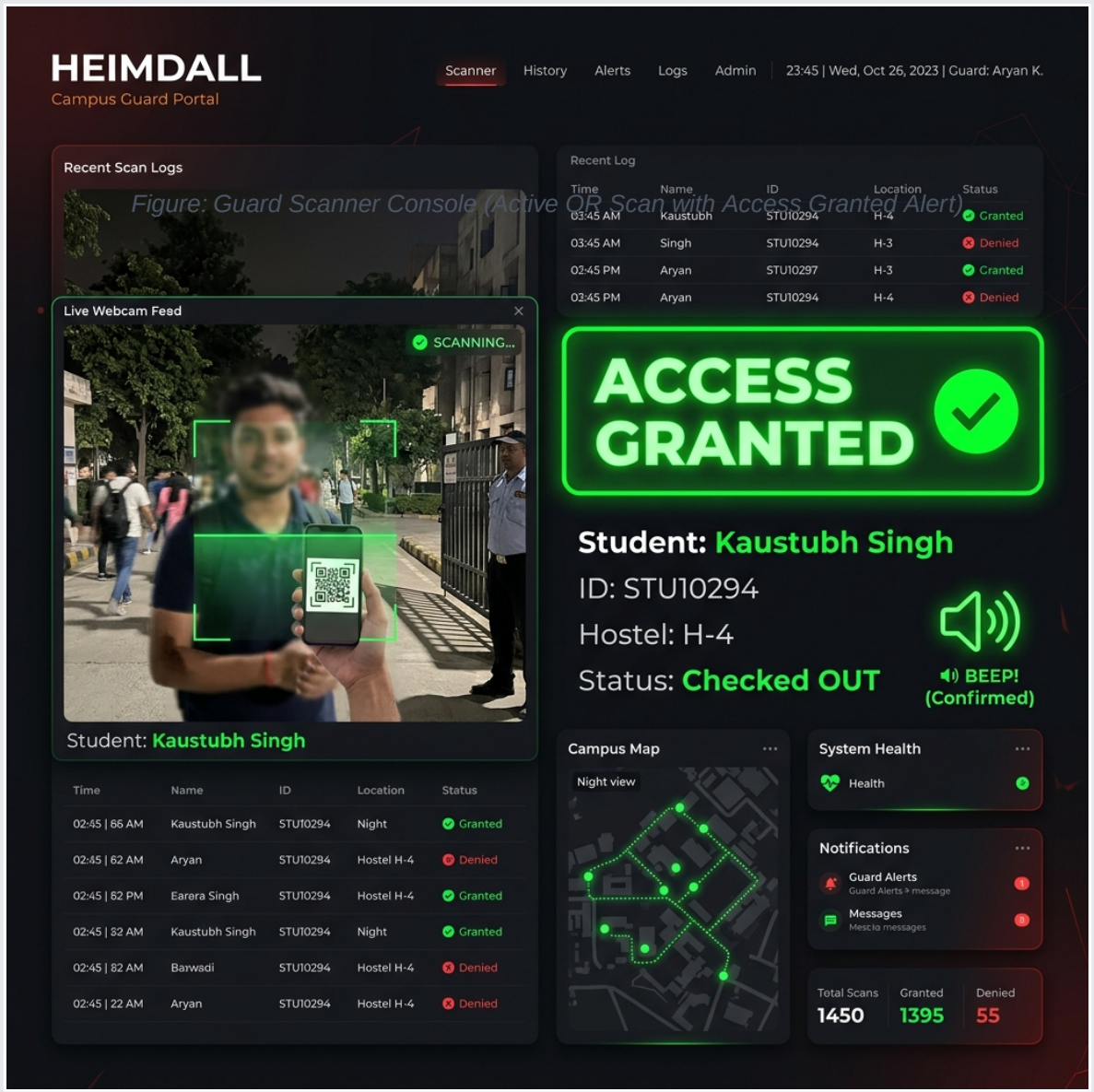


3. The Guard Scanner Portal

Deployed on terminals at campus entrance and exit gates, this portal allows guards to scan student QR codes in milliseconds. The scanner interface uses a fast native camera view and integrates socket-driven notifications to broadcast entry events immediately.

Detailed Operations Flow:

- **Dynamic Video Feed Scanner:** Connects directly to the guard's camera (desktop webcam or mobile lens) using a high-fps scanning engine.
- **Instant Alert Indicators:** Upon scanner capture, the system queries the Heimdall API. A successful scan renders a large neon green "ACCESS GRANTED" banner accompanied by an audible beep.
- **Access Denied Handling:** Displays a solid red warning when a QR code is expired, already scanned, or invalid. The system blocks double-scanning via an in-memory lock mechanism to prevent crowd gate-crashing.
- **Live Gate Feed:** Shows the last 5 scanned logs on the left sidebar in real-time, allowing security heads to review recently cleared students without loading logs screens.



4. The Warden Portal

The Warden Dashboard focuses on accountability, approvals, and student welfare. Wardens supervise outpasses, verify parental responses, and resolve grievances.

Detailed Approval Flow:

- **Queue Auditing:** Shows all pending home-visit applications. The dashboard aggregates student profile details, destination, and visit dates.
- **WhatsApp Status Tracking:** The portal displays real-time webhook status updates from parents. It clearly flags whether parent approval is "PENDING" (yellow warning), "APPROVED via WhatsApp" (green checkmark), or "REJECTED by Parent" (red error).
- **One-Click Override Actions:** Wardens can instantly "Grant Final Pass" to authorize the outpass, or "Reject Pass". This updates the database and releases a valid home-visit QR code directly to the student's portal.
- **Analytical Widgets:** Visual charts on weekly outpass volume, request status breakdown, and hostel distribution help wardens manage hostel capacity.



5. The Admin Analytics & Telemetry Portal

The Administrative Console is designed for IT managers and campus directors. It provides high-level telemetry, server performance metrics, and configuration controls.

Telemetry and Server Analytics:

- **Real-Time Traffic Graphs:** Charts active gate scans per minute to detect peak rush hours (e.g. 5:00 PM weekend returns).
- **System Health Gauges:** Displays CPU load and MongoDB connection count. Shows actual database connections (e.g., 5/100 connections) to ensure Atlas Free Tier caps are never exceeded.
- **Diagnostic Metrics:** Lists system uptime, active server versions, status of Redis caching, and error logs (zero-error status).
- **Active Campus Count:** Displays counts of students currently logged "IN CAMPUS" versus "OUT OF CAMPUS".



6. System Design & Security Safeguards

To support large campuses of 15,000+ students on modern hosting environments, Heimdall implements multiple defensive engineering architectures:

- **1. Anti-Duplicate Lock (`ScanLockService`):** An atomic locking service that blocks a student's QR code from being scanned twice within a 5-second window. This eliminates double-read database race conditions.
- **2. Free-Tier DB Optimizations:** MongoDB connection pool capped at 5 in free tier (`MONGODB\_TIER=free`) with 15-second idle cleanups. This leaves connection headroom for cloud monitoring.
- **3. Queue-Driven Notifications:** Parental WhatsApp notifications are handled in the background by BullMQ worker threads. If a WhatsApp API limit is hit, BullMQ automatically retries the job later without failing the gate scanner request.
- **4. Dynamic QR Encryption:** QR payloads include JWT structures containing user identifiers signed with an HS256 secret. QRs cannot be modified or forged by students.

Summary of Default Credentials for System Testing:

Role	Default Email	Password
Admin	admin@heimdall.com	admin123
Warden	warden@heimdall.com	warden123
Guard	guard@heimdall.com	guard123
Student	kaustubh@student.com	student123

